Formal Lab Report

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Unknown A

CHEM 2081 - 003 - 402

Introduction

The purpose of the formal lab report is to use nuclear magnetic resonance, gas chromatography-mass spectrometry, and fourier transform infrared spectroscopy images to identify an unknown compound. NMR, GCMS, and FTIR are tools that chemists use to analyze organic compounds. The graphs contain information about the structure and properties that chemists can use to identify an unknown compound or distinguish between two like compounds.

Body

Nuclear magnetic resonance spectroscopy is used to determine the location of functional groups that are present in a molecule. NMR is applied to a nucleus that has an odd mass number and/or odd atomic number. Radio frequencies are used to flip the nucleus of an atom to a higher energy state and then the energy is measured through observation of the signal decay that occurs over a period of time. A sample is placed in a long tube and then placed in a superconducting magnet for this to occur. Then a frequency is applied which causes the atoms to spin. As the spinning atoms slow, the nucleus flips back to the lower energy state, where is realigns itself with the magnetic field. Each signal observed is given with a differing strength, allowing for identification of molecules based their behavior in the magnetic field. The signal that is measured is recorded as an oscillating curve on a graph. The spectrum can be analyzed to find out the formula and structure of the sample.

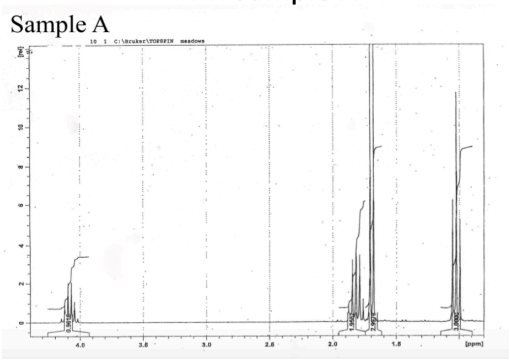
Fourier transform infrared spectroscopy uses the fact that when organic compounds absorb IR radiation, specific bonds within the compound begin to exhibit vibrational excitation. A sample is placed into the spectrophotometer where a beam of infrared light is separated into

different frequencies before it penetrates the sample. The molecules in the compound move in specific ways, and they require energy from different light frequencies to accomplish this. These energy differences required by the molecules to move in certain ways are reflected in their different IR absorptions, which are recorded on a graph that shows which light frequencies were absorbed by the sample. The graph is analyzed to determine the functional groups present in the unknown.

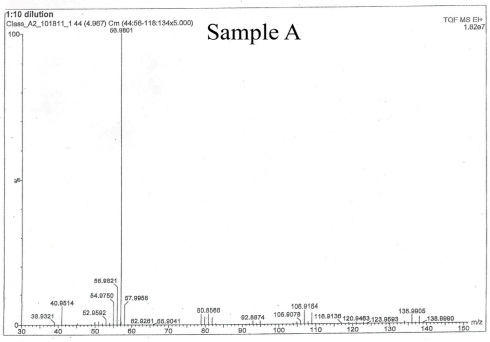
Gas chromatography-mass spectrometry utilizes mass to determine the identity of a sample. The gas chromatography separates the sample into its components in their gas phase, which are then exposed to an electron beam. After the components are exposed to the electron beam, they undergo ionization where they become charged particles. A magnetic field is introduced, which causes the charged particles to deflect according to their mass and focused on a detector. The mass spectrum is measured on a graph where the x-axis is the mass/charge ratio and the y-axis is relative abundance. The graph is analyzed to determine the identity of the components in the sample.

Data

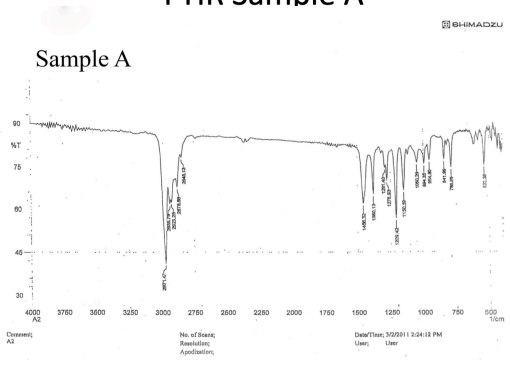
NMR Sample A



GCMS Sample A

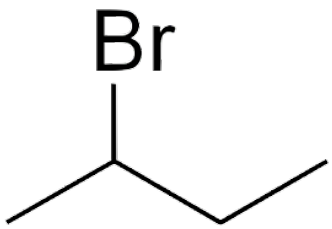


FTIR Sample A



Conclusion

There is a presence of four individual peaks in the NMR spectra that indicated four hydrogen atom environments. The simplest conclusion from this is that the basic structure is a 4-carbon chain. There is also presence of a doublet, which implies that there is 1nH. From the GCMS spectra, it is shown that the M+ has a mass of 138 g/mol and that the base peak at m/z 56.9, M-79, indicates a loss of bromine. From the FTIR spectra, there are three significant peaks. The largest peak has a frequency of 2971.74 1/cm and it falls within the range of an alkane. The second peak is at 1380.13 1/cm, which corresponds to another alkane. The third peak has a frequency of 532.38 1/cm, which corresponds to halogen, specifically bromine. The unknown structure was determined to be 2-bromobutane.



References

H McNeil. Extraction. Chem 2081 Organic Chemistry II Lab Manual, Macmillan Learning Curriculum Solution: Plymouth, 2023